



RUN4

OPERATION MANUAL

Four-track Tape & Texture Machine

Edition 1 — 2026

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PART 01

Getting started

What RUN4 is, how it is laid out, and the habits that make the rest of the manual read easily.

How this manual is written

- Controls are named as they appear on screen. T1 to T4 are the four track buttons; Tape, Poly, Disc and Perform are the four mode buttons in the top row.
- A tap is one short touch. A long-press is holding a control until its panel opens. Long-press is RUN4's route to assignment and detail throughout, and it never changes the value of the control you are holding.
- An arrow describes signal order, so Material -> Grain -> Tone -> Color -> Air -> mixer is the order audio actually travels in.
- Where material is a lookup rather than an explanation it is set as a table of trigger and action. The sentences around a table carry the detail that will not fit in a row.
- Every section here is also the in-app Help section of the same name. The app and this PDF are generated from one source, so they cannot disagree with each other.

What RUN4 is

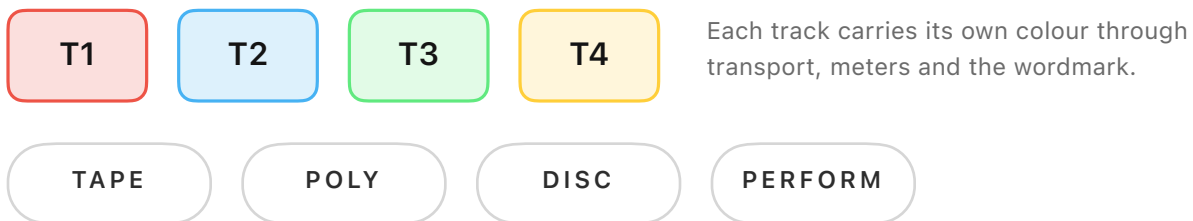
- RUN4 is a four-track Tape & Texture Machine. Each track has a Material source, a device chain, mixer controls, sends, and performance state.
- The normal path is: choose a track, choose the Material mode, load or capture sound, shape it with devices, perform it with transport/macros/scenes, then export a master or stems.
- The four main areas are Tape, Poly, Disc, and Performance. Tape, Poly, and Disc are sound engines. Performance is the full-screen live stage for four waveforms, hands-on track controls, Sets, Rune Pad, and Punch-in FX.
- The fixed device order is Material -> Grain -> Tone -> Color -> Air -> mixer/sends/master. Device defaults are transparent where possible, so moving a Mix or amount control is often what actually brings an effect into the sound.



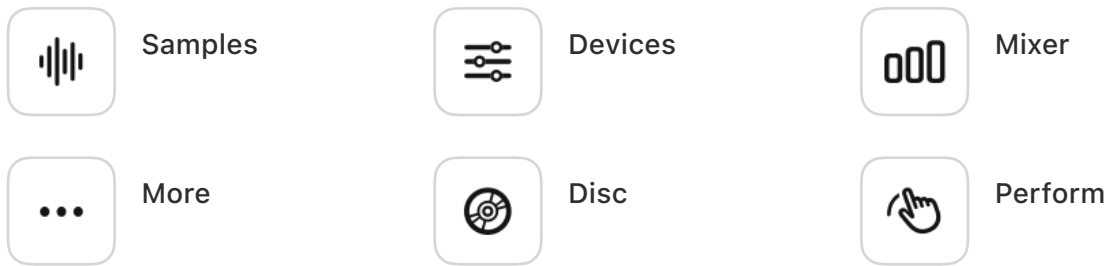
The fixed device order every track runs through

Navigation and tracks

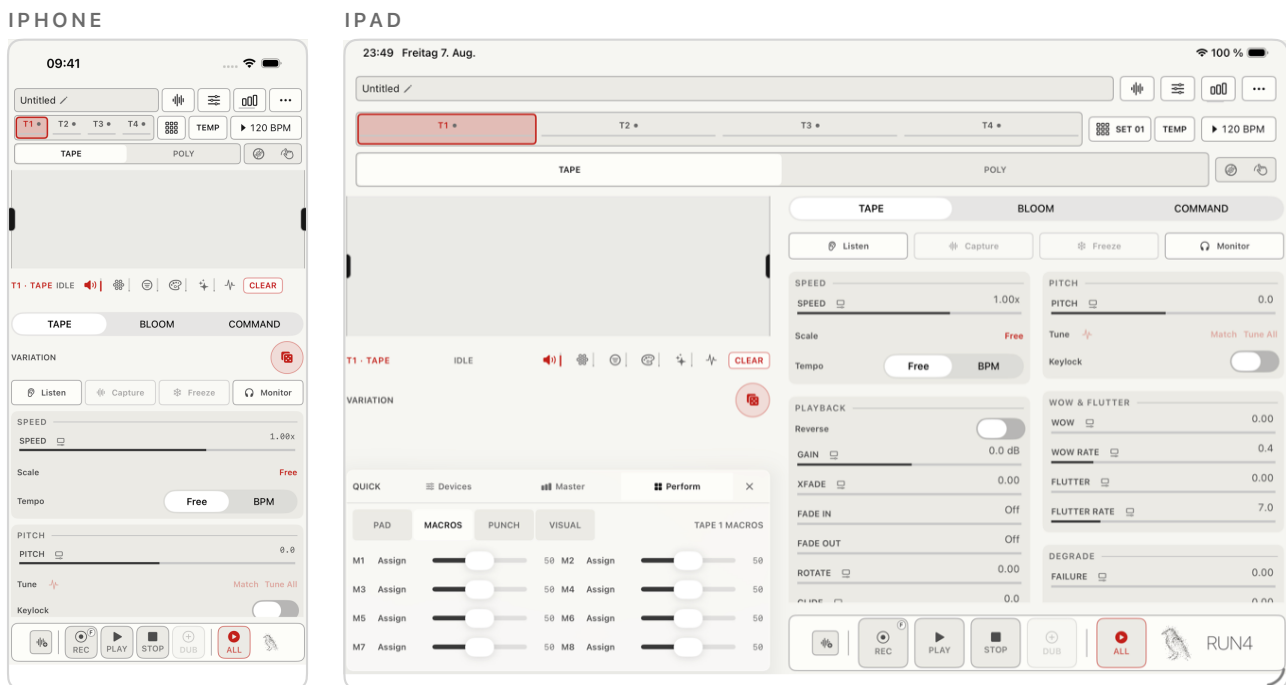
- T1 to T4 select the active track. Track color follows the selected track and appears in transport, device activity, meters, and the reactive RUN4 bird/wordmark.
- The top mode row selects the active view: Tape for recording/looping, Poly for keyboard/sample/synth/AUv3 playback, Disc for long streamed files, and Perform for performance structure.
- In the top row, the waveform button opens the Sample Browser, the sliders button opens Devices for the selected track, and the bar-chart button opens the Mixer. The ellipsis holds Projects, Settings, and Save.
- Inside Devices you reach Grain, Tone, Color, and Air for the selected track. Settings, which contains this manual, is in the ellipsis menu rather than on a gear of its own.
- The active device strip lights when a device has moved away from its clean default. Clear inside a device resets that device for the selected track.



Track selectors and the mode row



The top row, left to right after the project name, then the two workspaces. Disc and Perform sit outside the Tape/Poly control because they belong to the Set rather than to one track -- tap the hand to reach the Perform stage.



Tape on both idioms. The top row carries the project name, tools, T1 to T4, the tempo, and the Tape/Poly source control with the Disc and Perform workspaces beside it. iPad landscape lays the same controls out side by side instead of stacking them.

Projects and Auto Save

- Open the Project menu to Save, Save As New, create an empty project, load a project, or save the current setup as a reusable template. Save is always explicit and available whether Auto Save is on or off.
- Auto Save is on by default in Settings > General. It keeps the open project updated while you work, including when you leave RUN4.

- When Auto Save is off, RUN4 writes project files only when you choose Save or Save As New. Opening another project does not save the current one; starting a new project or using a template warns before unsaved changes are discarded.

Undo and Redo

- Destructive actions are checkpointed: pressing REC on a tape that already holds a take, Clear, the Reset choices, and replacing a loaded sample. Open the Undo menu in the top bar to step back. A restored checkpoint returns the sample, the loop in and out points, Pitch, and the rest of that tape's settings exactly as they were.
- Undo restores a whole project state rather than a single control, so stepping back also reverts changes made after that action. Redo brings them back: the state before each Undo is captured first, so moving backward and forward never loses anything.
- The history holds twelve destructive actions and survives normal editing and Auto Save. Opening another project or starting a new one clears it.
- DUB is deliberately not covered. An overdub layers into the existing take and rewrites its file in place, so material from before the overdub cannot be restored. Record onto another track when you want to keep the original.
- Smaller, local undos stay separate: the die in Tape, Bloom, and Command has its own one-step Undo per section, and the Tune row has its own revert buttons.

Long-press shortcuts

- Rec Grid quantizes Record, Overdub and Play All to RUN4's own tempo grid when no external clock is running; the Metronome clicks that same grid.
- When a parameter is assigned to one of the eight macros M1 to M8, its small M label appears beside the parameter name; tap that label to open the matching modulator directly.

LONG-PRESS TARGETS

BPM display, anywhere in RUN4	Timing & Sync: MIDI Clock In and Out, Ableton Link, Sync Quantize, Rec Grid, Metronome
Any routable parameter or selector	Assign or remove a modulation source, or map it to a track Macro or Rune Pad
Poly pattern step	That step's detailed editor
Quick device icon	That device's full panel

PART 02

Tape

The four tape tracks: recording, loops, tuning, Bloom, and the Command Sequencer.

Tape mode overview

- Tape records audio into a loop, plays it back, overdubs into it, and can resample the internal mix through Listen/Capture. Tape is the quickest mode for live sampling and transforming captured audio.
- The die in Tape, Bloom, and Command rolls a variation for the visible section. After a roll, a visible Undo button restores it. Undo is one step per section; a new roll or a manual change to that section starts a new state.
- REC records into the selected tape. PLAY starts playback using the Tape Start setting: Restart, Resume, or Align. STOP stops the selected tape. DUB overdubs onto existing tape material. ALL starts or stops all track transports together.
- Clear offers destructive reset choices. Only Tape clears the tape material/state. Tape + Devices also resets Grain, Tone, Color, and Air on that track.
- Listen arms the internal always-listening buffer. Capture copies the recent internal mix into the tape without needing to route speaker output into the microphone.
- Freeze holds the tape player at the current moment for sustained texture and performance gestures.



Rec



Play



Stop



Dub

Transport controls

Tape recording and fixed loops

- A normal tap on Tape REC uses the currently selected capture setup. Tap the timer inside the transport box to open its persistent capture panel.
- Record captures and stops at the end. Record & Play captures and immediately starts the new Tape as a loop.
- Length can be Free or a fixed 1, 2, 4, or 8 bars. Fixed length follows the current RUN4 tempo and 4/4 bar grid; for example, 2 bars at 120 BPM record for 4 seconds.
- The capture panel includes the Metronome and a continuous click-level fader. It remains open while you adjust the level. The click follows quarter notes, accents beat 1, and is never printed into the Tape or an export.
- Rec Grid and fixed capture length are related but separate: Rec Grid chooses when a transport action begins, while Length chooses when a fixed recording ends.

Tape parameters

- Listen/Capture keeps an 8-second internal mix window for quick resampling.
- Speed is varispeed playback from 0.0625x to 4x. It changes timing and pitch unless Keylock is enabled.
- Tempo switches Speed between Free behavior and BPM-aware behavior. In BPM mode, Tape BPM tells RUN4 what tempo the captured tape originally represents.
- Pitch transposes tape playback in semitones from -24 to +24.
- Scale chooses pitch snapping for tape speed/pitch behavior: Free, Chromatic, or scale-based choices. When Scale is active, Root chooses the tonal center.
- Start Grid and Length Grid appear in BPM mode. They quantize loop start and loop length to musical values from 8 bars down to 1/16, or Free.
- Keylock keeps pitch more stable while Speed changes. Reverse flips playback direction.
- Gain sets tape output level. Free waveform trims settle onto a nearby zero crossing, while a 3 ms safety blend protects hard source and tempo-grid edges without changing recording length. XFade extends that loop-wrap blend up to 500 ms, useful for pads and clean sustained loops. Rotate offsets the loop/read position. Glide smooths motion between playback changes.

- Fade In and Fade Out shape each loop pass and are Off by default: loaded or recorded material is never reshaped until you ask for it. Full range is half the loop, so both at maximum form a single triangular swell. Setting only one of the two still leaves the loop wrap click-free, because the opposite edge is given a short automatic minimum.
- The waveform draws Fade In and Fade Out at the length you set, and draws nothing when they are Off. Click-free looping is playback behaviour rather than an edit to your material, so it is not drawn.
- SOS controls sound-on-sound retention during overdub: high values keep previous material longer, low values replace it faster.
- Wow and Wow Rate add slower tape-speed drift. Flutter and Flutter Rate add faster instability.
- Tape Type is Off by default, so existing tapes keep their original sound. Reel, Sixties, 4-Track, Cassette, Video, and Dictaphone each add a calibrated medium response through bandwidth, dynamics, a subtle noise floor where appropriate, and their own quiet mechanical movement. Only Sixties, 4-Track, and Dictaphone add a touch of soft saturation; Reel, Cassette, and Video stay clean.
- Tape Type never rewrites the front-panel Wow, Flutter, Failure, Noise, or Splice values. Those remain your additive performance controls: choose a medium first, then add as much motion, wear, hiss, or damage as the moment needs.
- Degrade Failure adds wear, pitch snags, crinkles, and dropouts. Degrade Noise adds hiss. Noise Color changes the character of that hiss.
- The die beside Tape Variation creates a restrained new tape character: Speed, Pitch, playback movement, Wow, and Flutter change, while your source, gain, loop, routing, and project tempo stay untouched. Long-press that die to restore the exact pre-roll settings.

Tape Tune, Match, and Tune All

- The Tune row is for bringing recorded Tapes into a musical relationship. It is available when the selected track has a tape take; key analysis works from the current loop region and reports the audible key, including the current Tape pitch and non-Keylocked speed shift.
- Tap the waveform/analyze button to detect the selected Tape's key. The result is a guide, not a forced correction: sparse, noisy, or highly atonal recordings can return No key.

- Match analyzes the selected Tape and the other recorded Tapes, then offers pitch-match choices. Choosing one changes the selected Tape's Pitch and switches its pitch snapping to Chromatic; the audio file itself is not rewritten. Use the curved-arrow button to revert the latest Match on that Tape.
- Tapping Match again reopens the same list instead of analyzing a second time, so Best, Fifth, Darker, and Brighter can be tried against each other; the choice in effect is ticked, and picking another replaces it rather than stacking a second shift. Use the analyze button when you want a genuinely fresh analysis, for example after retrimming the loop.
- Tune All > To T1–T4 uses the chosen recorded Tape as the reference and applies the best pitch match to every other recorded Tape it can analyze. The circular curved-arrow button reverts the last Tune All gesture as one action, provided those affected tracks have not been changed since.

Tape Bloom

- Bloom turns one tape read head into a small cloud of six read heads. It is useful for chorused loops, smeared harmonies, clouds, and animated tape swarms.
- Spread separates the read heads around the tape. Motion animates the read-head positions. Stereo widens the heads across the field.
- Warble and Warble Rate add pitch/time instability to the Bloom cloud. Degrade adds tape wear specifically to Bloom behavior.
- Each Head has Direction, Level, Pitch, Speed, and Pan. Direction can be Stop, Fwd, or Rev. Pitch is in semitones. Speed changes that head's playback rate.
- Reset Global restores Bloom's shared controls. Reset Heads restores all six head-level settings.
- The die beside Bloom · All Heads turns Bloom on and creates one balanced six-head cloud, rather than randomizing each head into unrelated extremes. Long-press that die to restore the previous Bloom state, including every head.

Command Sequencer

- Command is a 16-step performance sequencer for tape actions. It can jump, reverse, retrigger, squeeze loops, stutter, gate, hold, nudge, spread Bloom, or trigger Memories.
- BPM is the shared master tempo. Editing it here changes the same project tempo used by Link, MIDI clock, and synced behavior.

-
- Pattern selects A, B, C, or D. Chain lets four pattern slots define a performance order. Seed Lock keeps random/probability behavior repeatable; New Seed creates a new variation.
 - Each step has a command, an optional value, and Probability. Probability at 1 always fires; lower values make the step conditional.
 - Jump uses Position. Rate uses Rate. Spread changes Bloom spread. Squeeze changes loop size. Stutter uses Repeats. Nudge uses Offset.
 - Memory steps choose a stored Memory and can pitch it chromatically from -24 to +24 semitones. Level, Fade In, and Fade Out shape each triggered snippet. A Memory used by a Command step stays protected from automatic Memory-bank eviction. Clear Pattern resets the active pattern.
 - The die beside Command · Pattern fills the selected pattern with sparse, playable tape gestures. It never creates Memory steps, so it cannot unexpectedly trigger stored audio. Long-press that die to restore the entire previous selected pattern.

PART 03

Poly and Disc

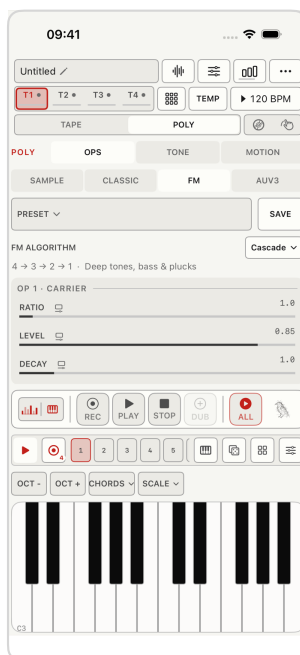
Keyboard, sample and synth playback, and long streamed files.

Poly mode overview

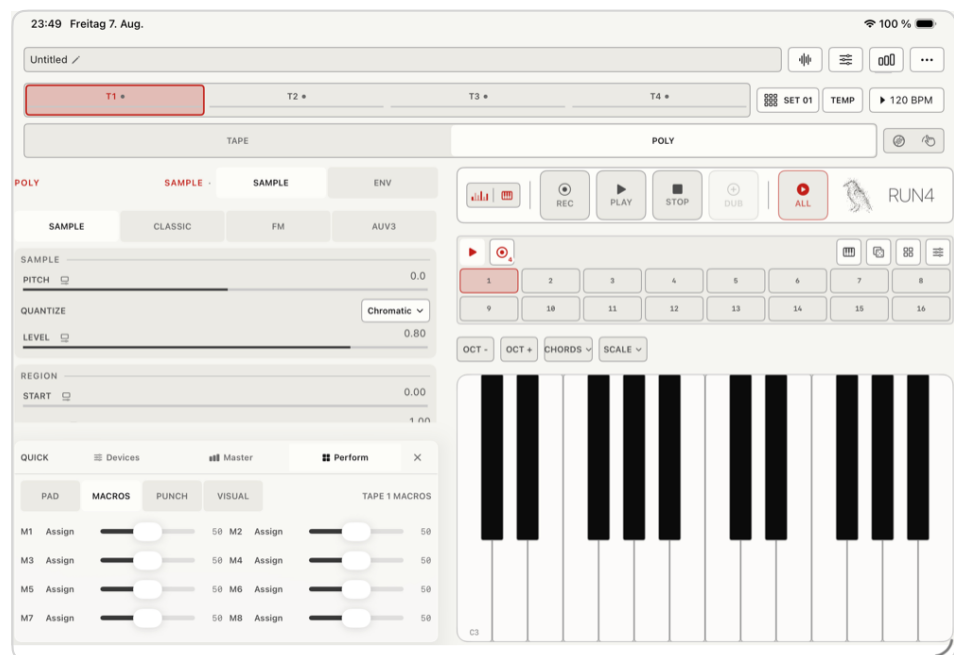
- Poly plays the selected track as a polyphonic instrument. It can use a loaded sample, the internal synth, or a hosted AUv3 instrument in the standalone app.
- The keyboard and MIDI share the same voice path. Octave buttons shift the playable range; the current base octave is shown at the bottom of its C key on the keyboard.
- Choose Sample, Classic, FM, or AUv3 directly at the top of Poly. Classic and FM are RUN4's internal synth engines; in the AUv3 extension build, AUv3 hosting is hidden because iOS does not allow an AUv3 extension to host other AUv3 instruments.
- The keyboard and step sequencer can be shown or hidden independently with the two display buttons inside the Poly transport box. Hiding them also collapses their keyboard/scale controls and returns the freed space to the selected Sample, Synth, or AUv3 editor.
- OCT -, OCT +, CHORDS, and SCALE share one control row. Tap CHORDS or SCALE to open its menu.
- From T1 to T4 selects a Tape as the musical reference. Match analyzes that Tape and applies its detected root and major/minor key to the active keyboard mode: SCALE receives a matching guide, while CHORDS receives the corresponding diatonic chord map. It does not change the source Tape. The CHORDS and SCALE menus can lock performance input to the active chord map or matched scale. A lock keeps the piano layout intact, dims unavailable keys, and filters live MIDI/QWERTY notes as well as on-screen playing.
- Poly's 16-step strip is a compact note or kit input for the selected Poly source and for printing ideas into Tape. It does not arrange Tape tracks or build songs and is intentionally not a complete song or multitrack sequencer.
- Tap a step to switch it on or off. Long-press a step to open Velocity, Gate, Probability, Nudge, sample pitch, and kit controls. The pattern settings menu sets length, resolution, Step Entry, and Auto Advance.

- With Step Entry on, played MIDI, onscreen, or QWERTY notes remain audible and are written into the selected step; notes played together become a chord. Auto Advance moves the selected step after that entry. The record-circle prints Poly to Tape; its menu selects Free or a fixed bar length. The dice randomizes performance values.
- A Poly Kit holds eight sample slots for a playable drum or multi-sample setup. In Samples, choose Load Poly Kit, then choose the destination track and one of its eight slots. Poly Kit Presets save and recall those eight sample-slot choices only, so the same kit can be used with different patterns; patterns remain separate until a future Kit + Pattern preset is deliberately created.
- Sample, Synth, and hosted AUv3 instruments can be performed or printed directly into the selected Tape. Fixed-bar printing is useful for tempo-aware AUv3 grooveboxes because RUN4 supplies the project transport and closes the Tape on the selected bar boundary.

IPHONE



IPAD



Poly on both idioms. Keyboard, pattern, and sample or synth controls for the selected track; on iPad the keyboard and the pattern grid sit beside the parameters rather than below them.

Poly sample controls

- Pitch sets the sample pitch offset in semitones. Pitch Quantize chooses Chromatic semitone snapping or Free continuous detune.

- Start chooses the sample start point. Length chooses how much of the sample is playable from Start.
- Level sets per-voice output level. Loop chooses the loop start point inside the playable region. XFade smooths the sample loop transition for up to 500 ms.
- Loop Mode Fwd loops forward from end back to loop start. Flip ping-pongs forward and backward.
- Attack, Decay, Sustain, and Curve shape the amp envelope. Decay currently also acts as release for the sample voice.
- Velocity controls how much MIDI or keyboard velocity affects loudness.
- Fltr Amount applies a filter envelope to the voice. Fltr Shape changes how that filter envelope follows or differs from the amp envelope.

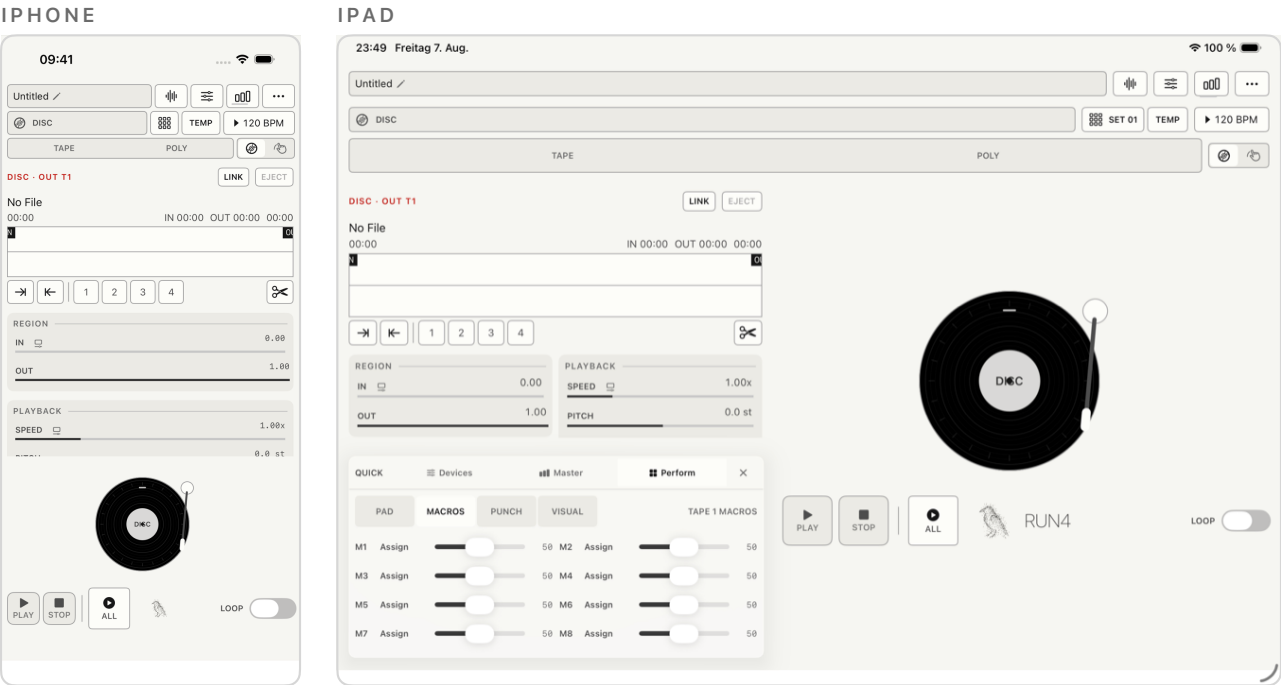
Poly synth and AUv3

- Synth mode offers the Classic subtractive engine and the four-operator FM engine. Presets are scoped to the selected engine and can be selected or saved.
- Classic and FM organize their controls into named sound, tone, and motion sections. FM starts in Ops, where its four operator levels and algorithm are most immediate; Sample region and loop controls are only shown when Sample is selected.
- In Classic, Waveform chooses Sine, Triangle, Saw, or Square. In FM, Cascade, Merge, Dual, and Split select the operator routing.
- Synth Filter has Cutoff and Resonance. Cutoff is mapped musically from low to high frequency; Resonance adds emphasis without intentional self-oscillation.
- Amp Env has Attack, Decay, Sustain, and Release. Filter Env has Env Amount plus Flt Attack, Flt Decay, Flt Sustain, and Flt Release.
- LFO has Shape, Rate, Pitch Amount, and Cutoff Amt. Pitch Amount reaches up to about an octave of peak pitch movement; Cutoff Amt moves the filter over several octaves.
- AUv3 mode scans available Music Device instruments. Select loads an instrument, Edit Plugin opens its editor when available, and Remove AUv3 unloads it.
- Hosted AUv3 instruments receive RUN4's tempo, beat position, time signature, and play/stop state. This lets transport-aware grooveboxes and sequencers start, stop, and print against the same musical grid.

- When supported by the plugin, its editor can open Docked inside Poly, Compact, or Full Screen. RUN4 keeps the selected Poly page and editor presentation when the device rotates.

Disc mode

- Disc is one shared performance deck for the Set. It streams a long file from storage instead of loading it into the tape/sample buffer.
- The top T1 to T4 buttons choose the active working track. Disc's Output menu separately chooses where the shared deck is heard: Follow Selected moves with the active track, while Pinned keeps it on a fixed track.
- OUT T1 to OUT T4 shows the effective route and its track color. Tap that badge to select the routed track and edit the Grain, Tone, Color, Air, sends, and mixer channel processing Disc.
- Link opens file selection/loading for the shared Disc source. The displayed filename confirms the current file. Eject removes only RUN4's link to that file — it never deletes the original file from Files, iCloud, or an external drive. Position can be scrubbed in the position bar.
- Level sets Disc output. Start chooses the normalized file start point. Loop repeats the file/region when enabled.
- Disc has its own PLAY and STOP plus the same ALL behavior as the other transport surfaces.



Disc on both idioms. A global workspace for long streamed files, outside the four tracks.

PART 04

Devices

The fixed device chain every track runs through: Grain, Tone, Color, Air.

Grain device

- Grain is an insert device that turns the source into grains, clouds, scans, and delay-like textures. Mix defaults to 0, so the device is clean until you blend it in.
- Page 1: Pitch transposes grains. Scale snaps Pitch to Free, Chromatic, Major, Minor, Dorian, or Pentatonic. Rate controls grain spawn density. Size controls grain duration.
- Page 1: Contour changes the grain window from sharper to softer. Warp moves or bends the read position. Spray adds position randomization and stereo spread. Pattern moves from rhythmic behavior toward cloud behavior. Scan reads through the recent buffer/history. Mix blends dry source and grains.
- Page 2: Detune randomizes grain pitch. Random Rate jitters spawn timing. Random Size jitters duration. SOS controls feedback/retention in the grain buffer.
- Page 3: Engine selects Live Cloud or Scan Cloud. Scan Mode chooses Stretch, Position, Delay Sync, or Delay Free. Feedback chooses SOS or Grain Output. Rand Pitch chooses Detune, Random, or Random Mono. Spray Mode chooses Warp or Warp Mono.
- Page 3: Buffer Lock holds the current grain buffer. Run While Idle keeps the mosaic engine moving even when the source is idle. Save Buffer stores the current granular buffer as reusable material.
- Page 3: Rand Amp randomizes grain amplitude. Rand Rev adds reverse probability. Stretch slows the scan read head. Delay Time and Delay Beats set free or tempo-synced delay-style scan timing.

Tone device

- Tone is the filter/resonator device. It defaults transparent: lowpass is fully open and Mix is 0 until you engage it.
- Type chooses Lowpass, Highpass, Bandpass, Formant, or Resonator.

- For Lowpass/Highpass/Bandpass/Formant, Cutoff sets the corner or vowel position, Resonance emphasizes the filter, and Mix blends the filter into the track.
- For Resonator, Voices chooses 1 to 16 resonant bands. Root shifts the tuned bank. Structure moves from harmonic to stretched/metallic spacing. Resonance changes ringing/decay. Brightness tilts the bank. Scale can quantize resonance tuning to musical scales. Mix blends the resonator.

Color device

- Color adds drive, reduction, and modulation character. It defaults transparent with Mix at 0.
- Type chooses Saturate, Fold, Bitcrush, or Ringmod.
- Saturate is warm tanh-style saturation. Fold creates richer polynomial wavefolding harmonics. Both use Drive, Tone, and Mix.
- Bitcrush uses Bits and Rate. Lower Bits reduces bit depth; lower Rate reduces sample rate. Tone darkens or opens the result. Mix blends it in.
- Ringmod uses Drive, Frequency, Tone, and Mix. Frequency sets the carrier from low tremolo/ring tones up into metallic sidebands.

Air device

- Air is the space device: reverb, delay, freeze, repeat color, and broad tonal shaping.
- Freeze holds the reverb tail. While frozen, the tail sustains as a performance layer.
- Reverb Type chooses Room, Hall, Plate, or Space. Air Mix controls reverb level. Air Size controls tail size/feedback. Damping darkens the reverb.
- Delay Type chooses Digital, Ping-Pong, Tape, or Diffusion. Delay Mix controls delay level. Delay Time sets repeat timing. Delay Feedback controls repeat regeneration.
- Delay Pitch shifts repeats up or down. Chromatic snaps delay pitch motion to semitones; Glide leaves it continuous. Repeat Tone darkens repeats. Saturation soft-clips the delay feedback path.
- Tone is a broad bipolar space tone: negative leans darker/low-pass, positive leans brighter/high-pass.

PART 05

Sound library

Getting material in, and keeping the states worth returning to.

Sample Browser

- The folder button opens Samples. It contains Memories, Recordings, Recently Imported files, and Linked Folders.
- Import copies or links selected audio files through the system picker. Link Folder adds a browsable folder tree using security-scoped access. AudioShare opens AudioShare and returns imported audio to RUN4 when available.
- Tap or the play icon auditions a sample. Stop ends audition. Load Tape loads the selected item into Tape. Load Poly loads it into Poly. Load Disc links it for Disc streaming.
- Recordings can be deleted. Imported history entries can be removed. Samples can be renamed. Linked folders can be unlinked. Missing or unsupported files are marked in the browser.

Memories

- Memories are short captured moments of the overall mix. They are stored as reusable audio snippets and appear in the Samples browser.
- Memory Listen is a single global capture switch in the Memories section. It captures the post-master mix without loading into or changing a Tape. Tape Listen and Memory Listen share this one internal capture tap: turning Memory Listen on turns any armed Tape Listen off.
- Remember stores the most recent capture window. Auto stores a Memory every 8 seconds while Memory Listen is armed. If the capture buffer has not filled with audio yet, RUN4 reports that instead of creating an empty Memory.
- A Memory can be auditioned, recalled to Tape, or recalled as a variation: Reversed, Pitch +7, Octave down, or Half tempo.

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- The bank is bounded: older unreferenced Memories roll off when full, while a Memory used by any Command Sequence step is kept. Command Memory steps can trigger stored Memories from the tape sequencer and pitch them chromatically.

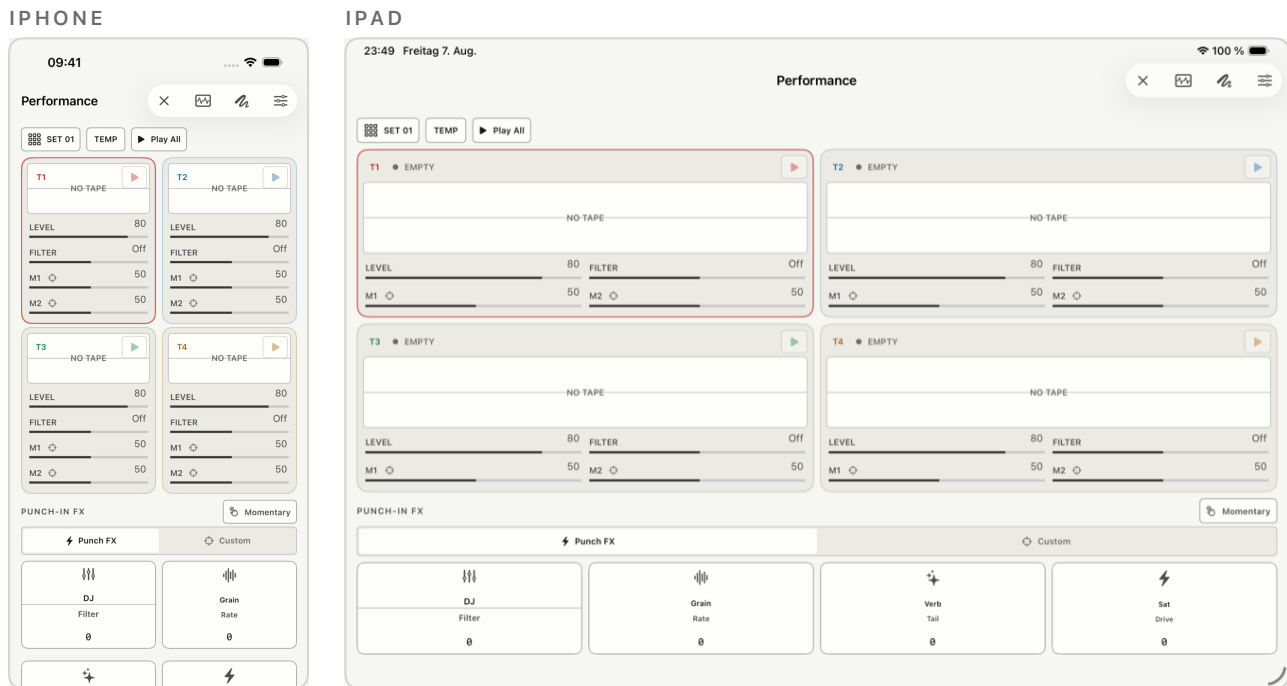
PART 06

Performance and visualizer

The live stage: Sets, Rune Pad, macros, and the stage graphics.

Performance mode, Sets, Rune Pad, and macros

- Performance opens the dedicated live workspace on iPhone, iPad, and Mac. Closing it returns directly to the Tape or Poly view you came from.
- Its four fixed track lanes show T1–T4 waveforms with per-track Play/Stop, Level, DJ Filter, M1, and M2. Keeping every Tape in a stable position makes live switching predictable; phone portrait uses a scrollable stage so the controls retain useful touch length.
- The top strip recalls Sets, toggles TEMP, and starts or stops all tracks. A Set stores the four tracks, material/device state, mixer state, macros, links, and performance setup. Open the Set menu to recall a slot, store the current state, or rename the current Set.
- TEMP is a safe performance layer for temporary parameter changes. Leaving TEMP restores the stored Set; it does not remove an AUv3 instrument or effect.
- Punch-in FX switches between RUN4's immediate global effects and assignable Custom slots. Momentary responds only while a pad is held; Hold latches the effect until it is released.
- The Rune Pad button reveals the XY performance surface and its grouped mapping controls. It appears inline on iPhone and portrait layouts, and as an overlay in wide landscape so the four track lanes remain visible.
- Performance Settings separates Macros and Track Links into dedicated pages. Macros M1 and onward can target parameters from grouped track/device options; each macro has a Value control from 0 to 1 and an assign menu.
- Macro CC learn maps incoming MIDI CC to a macro. For direct mapping, long-press a routable parameter: Modulation and Macro choices stay first, while MIDI Learn sits at the bottom. RUN4 reuses that parameter's assigned macro or a free macro slot, then learns the next incoming CC. Context menu Clear MIDI CC removes the learned assignment.
- Track Links create reactions between tracks. A source track can point at a target track, choose a link character such as ducking, and set intensity.

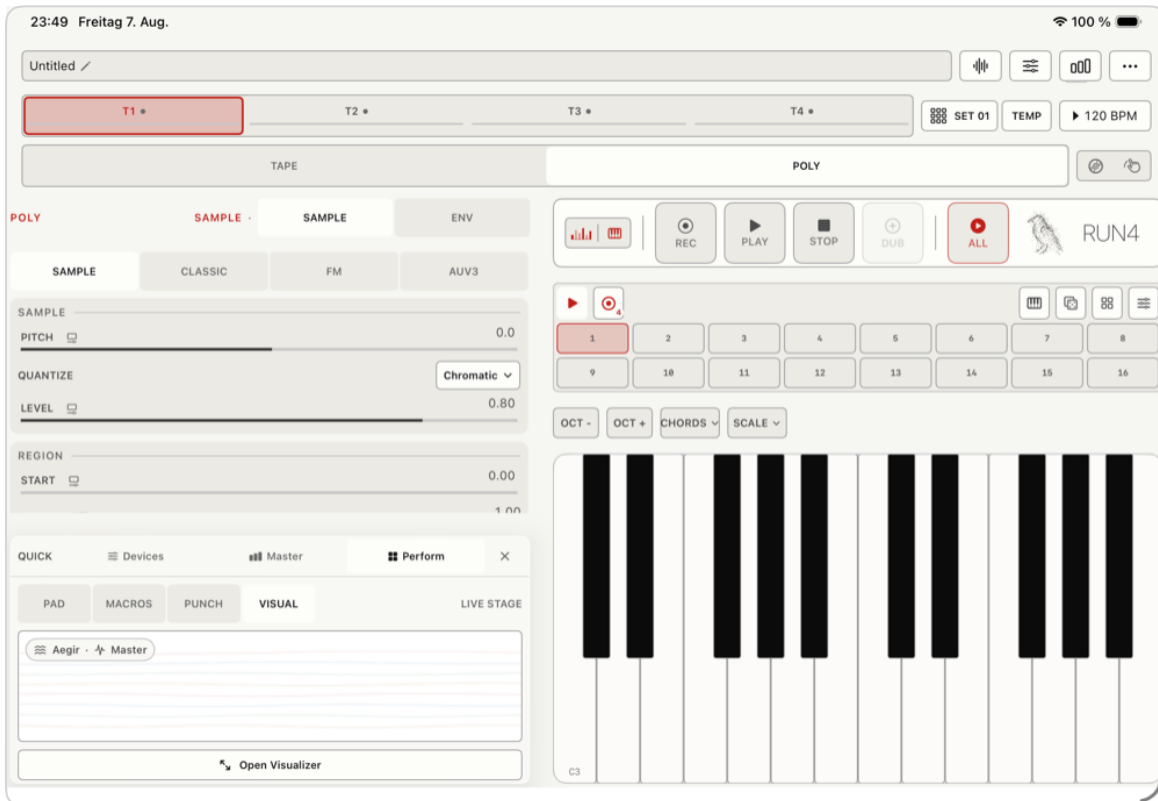


Performance on both idioms. Four track panels with level, filter and macros, over the Punch-in FX row.

Visualizer

- The visualizer turns the four tapes and the master signal into stage graphics. It is read-only: it never changes transport, routing, or any parameter, and it is not part of the AUv3 plug-in.
- It opens in two places. In Performance, the waveform button in the top toolbar fills the display. In iPad landscape, the Quick dock has a Visual tab holding a live preview; Open Visualizer runs the full stage inside the dock, so tapes, mixer, and transport stay visible and usable around it.
- From the docked stage, Full Screen promotes the same scene to the whole display. The docked stage stays underneath, so closing full screen returns to the dock you came from, with scene, palette, and source unchanged.
- Tap once to show or hide the controls. They hide again after about two seconds. While they are visible you can Close, go Full Screen, choose the audio source, choose the scene, switch palette, and open Touch FX.
- Source selects what drives the graphics: the master signal, or a single tape. Choosing one tape parks the others so the composition stays legible, and that tape's own spectrum drives the scene.

- Palette switches between Tapes, which uses the four track colours, and Mono, which uses a single hue and lets brightness carry the information.
- Swipe left or right to change scene. With Touch FX enabled, a one-finger drag belongs to the effects, so scene changes move to a two-finger swipe; the scene menu in the controls always works either way.
- Touch FX makes the stage a second performance surface. It reuses the Rune Pad engine, so a finger on the stage drives the same effects the Rune Pad does. It is off by default and configured from the Touch FX button.
- Scenes react to the frequency content of the material, not only its level, so bass, body, and treble drive different parts of a picture. Reduce Motion simplifies every scene: fewer elements, no drift, and no flicker.



The Quick dock on iPad landscape, with the Visual tab holding a live preview. Open Visualizer runs the full stage inside the dock, so tapes, mixer and transport stay usable around it.

Visualizer scenes

- Draupnir. Tape filaments radiating from a common centre, opening and bending with each track.
- Elivagar. Particle streams along curved trajectories sharing one vanishing point.
- Jormungandr. Arcs and points orbiting a centre, tangling as the material builds.
- Aegir. A slow ocean surface.
- Yggdrasil. A tree whose trunk takes weight from the low bands and whose canopy fills out with the highs.
- Jotunheim. Mushroom colonies that emerge, mature, and decay; bass material grows the field.
- Idavoll. Four clusters, one per tape.
- Urd. An hourglass of falling particles.
- Ran. A folded wave mesh.
- Vetrarbraut. A star road running to a distant point.
- Nornir. Points joined into shifting constellations.
- Huginn. The RUN4 bird.
- Galdr. Rune script drawn and redrawn.
- Midgard. A spectrogram landscape: frequency across, time receding to a horizon, level as height.
- Thor. A cloud bank with one tape per layer, and lightning that strikes more often as the material gets louder. The bolt takes the colour of the loudest tape.
- Bifrost. The rainbow bridge, built from the four tape colours blended into a ramp. Each stripe carries one frequency band.
- Valhalla. A sheet of birds in three depth layers, each bird riding its own part of the spectrum.
- Skidbladnir. A longship on the swell. The sail's four panels are the four tapes, the shields along the hull follow each track's level, and the hull takes its pitch from the wave beneath it.
- Langedr. The long-fire of a hall. Four logs, one per tape, glow with their own track; the mids are the flame body and the highs throw sparks. The embers keep breathing when the stage is silent.

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- Niflheim. Frost creeping in from the edges of the glass. The highs grow the crystals, the low end thaws them back, and each seed belongs to one tape.
 - Ginnungagap. The gap where the realms meet: ice reaching in from one side on the highs, fire from the other on the lows, and a churning seam that moves toward whichever side is winning.
 - Nordrljos. Northern lights over a ridge. One curtain per tape, each hung from its own baseline and rippled along its length by the spectrum.

PART 07

Mixing and routing

Levels, sends, the master chain, hardware routing, and getting audio out.

Mixer, sends, and master

- The Master panel shows master level, master dynamics, per-track channels, four sends, and AUv3 effect slots. On a wide landscape canvas, Mixer and the selected Send Bus share the screen equally so the console stays focused and avoids a delivery section below the fold.
- Each track has level, pan, and DJ filter controls. Meters show track and master signal.
- Send A through Send D route every track into four independent send buses. Each selected bus shows its four long send faders, return level, native effect, and AUv3 chain together in one signal-flow block.
- BUS FX inside a track's Devices window is a live view of those same shared A–D buses, not a duplicate plugin chain. The Send belongs to the selected track; Return and FX belong to the shared bus and therefore affect every track feeding it.
- Long-press a BUS FX Send or Return to assign modulation, a Macro, or Rune Pad control. AUv3 bus editors can open Docked directly below the routing/FX cards, Compact, or Full Screen.
- Master dynamics include compressor and limiter controls for final level management before export.
- AUv3 effects can be loaded directly on the selected bus or on the Master chain, edited when they provide an editor, bypassed, or removed.

Audio interface routing

- Open Settings > Audio > Input & Output Routing to configure a connected class-compliant USB audio interface. RUN4 stores the assignments, but validates them again whenever the hardware route changes.

- Link Audio Inputs in the same routing screen lists audio channels published by compatible Link peers. Enable Link and Audio Sharing, then assign any discovered channel independently to T1–T4. Record and Record & Play capture the selected network stream using the same Free or fixed-Bar length controls as hardware input. The Tape Monitor control listens to the selected Link source live; otherwise it monitors that Tape's hardware input.
- Hardware Route leaves that Tape on its physical input assignment. If a saved Link peer disappears, RUN4 preserves its name for reconnection and safely uses the hardware route until the channel returns.
- Tape Inputs assigns an independent mono channel or stereo pair to T1–T4. This allows simultaneous captures such as input 1 to T1, input 2 to T2, input 3 to T3, and input 4 to T4. Detect Input Channels briefly opens the record-capable route when iOS does not expose the full input count while idle.
- Hardware Outputs sends each Tape either to Master, Off, a physical mono output, or a physical stereo pair. Master Output can stay on the normal system stereo pair, move to another physical pair, become mono, or be disabled.
- A Tape assigned directly to hardware is removed from the dry Master path so it is not heard twice. Its send-bus contribution still returns through Master, keeping shared effects on the main mix.
- Examples: route T1–T4 to mono outputs 1–4 for a four-channel mixer, or to stereo pairs 1–2, 3–4, 5–6, and 7–8 for four independent stereo stems.
- If an assignment is invalid, the interface disconnects, or the multi-output renderer cannot start safely, RUN4 restores the normal stereo Master automatically. Physical channel counts and multi-output availability depend on the connected interface and iOS.

Export and bounce

- Open Export & Bounce from the top ... menu. It is a separate full workspace so delivery controls do not lengthen or clutter the live mixer.
- Export records the live output in real time. Select Master and/or T1 to T4 Stem, then choose Record or Record & Play. Stop Export finishes and opens sharing for the created files.
- Offline Bounce renders without playing in real time. Select Master and/or supported track stems, choose Duration, then Bounce Offline.

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- Offline stem support depends on the source and state. The app shows a detail line for unsupported options instead of silently failing.
 - Use live export for performance captures and gestures. Use offline bounce for fixed-length renders, clean stems, and transfer to another DAW.

PART 08

Sync and control

Playing RUN4 from, and alongside, everything else.

MIDI settings

- Input chooses All MIDI sources or a specific source. Refresh Devices rescans available sources. Bluetooth MIDI opens the iOS Bluetooth MIDI connection sheet.
- Poly Channels assigns a separate MIDI channel to each of T1 to T4 for note input. Control Channel chooses which channel receives global controls, or All.
- Program Change enables MIDI program changes for Set recall. Macro CC enables MIDI CC control of learned macros.
- MIDI Mappings holds a Transport section above the macro banks: press Learn next to Play All, Stop All, Record, or Dub and then send the button or pad you want to use. Learn accepts a note or a CC, so pad controllers work as well as dedicated transport buttons, and a pad mapped here drives transport instead of playing a note. Record and Dub follow the selected track and are ignored on a Disc deck. Unlike macros, these mappings belong to the controller rather than the project, so they stay put when you switch projects. The section's toggle disables controller transport without deleting what you learned.
- Save Current as Layout stores the controller's transport buttons and Macro CC positions for reuse across projects. Macro targets stay with each project, so one controller layout can play different project-specific macro assignments.
- MIDI Clock In/Out, Ableton Link, Sync Quantize, Rec Grid, and the Metronome all live in Timing & Sync -- open it here or long-press the BPM display anywhere in RUN4. MIDI Clock In lets RUN4 follow external MIDI clock and transport; MIDI Clock Out sends RUN4 clock to other devices; Sync Quantize queues synced actions (Record, Overdub, Play All) to the next beat while Link or MIDI Clock is running.
- Rec Grid quantizes Record, Overdub, and Play All to RUN4's own tempo grid when neither Link nor MIDI Clock is running -- pick a musical value (8 bars down to 1/16) and a punched action waits for the next grid line before it starts; Free launches immediately, as before. The first take defines the grid, so later records on other tracks lock to it.

- The timer inside Tape's transport box opens the capture panel for Record or Record & Play, Free or a fixed 1/2/4/8-bar length, the Metronome, and its continuous level fader. Fixed captures close automatically; Record & Play immediately loops the new take.
- The Metronome clicks that same grid: quarter notes at the project tempo, accenting beat 1 of each bar, so a quantized record has an audible reference. It runs on its own path and is never captured into a recording or export. Timing & Sync provides its continuous level control.
- Virtual Input exposes RUN4 as a MIDI input endpoint for other apps on the device.

Ableton Link and AUv3

- Project BPM is available from the top mode row. Drag the BPM chip to change tempo, tap it to switch the same compact chip to fine +/- 0.1 steps, or long-press it to open Timing & Sync.
- Ableton Link syncs tempo and transport with other Link-enabled apps on the local network. Timing & Sync shows Link status and transport state.
- RUN4 updates Link tempo from the project BPM and can react to Link start/stop when enabled in the Link settings sheet.
- When Link Audio is enabled, RUN4 can share the master Mix plus four post-fader track streams with compatible Link Audio peers that request them on the same local network.
- RUN4 can also receive compatible Link Audio channels directly into any Tape. Choose the source under Settings > Audio > Input & Output Routing; the normal Tape recording controls then capture that stream without an AUv3 routing utility. Tap Monitor on that Tape to hear the assigned Link channel live without recording it twice.
- RUN4 can host AUv3 instruments in Poly and AUv3 effects in the mixer/master path.
- Hosted AUv3 instruments and effects receive RUN4's tempo, musical position, time signature, and transport start/stop. Tempo-driven plugins such as gates, filters, and grooveboxes can therefore follow the same RUN4 transport used for Tape and fixed-bar recording.
- AUv3 editor choices are Docked, Compact, and Full Screen where supported. Docked keeps the plugin in the current RUN4 workspace; rotation preserves the selected device tab instead of returning to the first page.

- RUN4 can also run inside compatible hosts as AUv3 FX, Source, and Instrument components. Some standalone-only features, such as hosting another AUv3 instrument, are hidden inside the extension.

Mac and AUv3 editor note

- RUN4 is the full native app on iPad, iPhone, and Apple-silicon Mac. The standalone Mac window uses the complete RUN4 interface and can be resized freely; the layout follows the real window size.
- RUN4 AUv3 uses that same complete interface — Settings, playback modes, Scenes, Tape, Bloom, Poly and Command — without a reduced plug-in view or a second zoom overlay.
- An AU host owns the embedded editor window and its resize handles. RUN4 asks it for the largest available editor viewport, then adapts the real interface to every size the host provides.

PART 09

Reference

Working habits, key commands, and what to do when something misbehaves.

Performance tips

- For fast resampling, enable Listen, perform across tracks, then Capture into an empty Tape. This avoids microphone bleed and keeps the capture internal.
- For evolving pads, use Tape with Bloom, Grain Mix, Tone Resonator, and Air Freeze. Keep Color subtle unless you want obvious grit.
- For rhythmic glitch, use Command with Jump, Stutter, Gate, Retrigger, and Nudge. Use Probability below 1 for variation.
- For playable instruments, load a short sample into Poly, set Start/Length/Loop/XFade, then shape Attack/Decay/Sustain and play by MIDI.
- For long backing layers, use Disc instead of Tape so the file streams from storage and does not need to live in the tape buffer.
- For mix clarity, keep device Mix controls low at first, then raise only the effect that contributes to the part.

Keyboard shortcuts

- Connect a physical keyboard to use Poly as a computer MIDI keyboard. In Poly, press M or tap Keys to turn QWERTY MIDI on or off. Keys On is shown while performance keys are active.
- Command- and Control-modified keys keep their normal system and text-editing behavior. Leave a text field before performing so its typing focus does not take the keyboard.

WITH QWERTY MIDI ON

A W S E D F T G Y H U J	One chromatic octave from Poly's Base note
K	
Z / X	Octave down / octave up

Shift + key	Stronger note velocity
M	Turn QWERTY MIDI off again
WITH QWERTY MIDI OFF	
1 to 4	Select track
[/]	Previous / next track
T / P / D / F	Tape / Poly / Disc / Perform
R	Start or stop the selected Poly pattern while Poly is active
Space	Start or stop global playback

Troubleshooting

- No sound: check selected track, transport state, Material mode, tape/sample/disc source, track level, master level, and whether device Mix controls are unexpectedly at 0 or 1.
- Recorded over a tape by mistake: open the Undo menu in the top bar and choose Undo Record. The previous take's audio also remains in the Sample Browser under recordings, so it can be reloaded manually even later.
- Capture is silent: enable Listen first, wait for the listen buffer to contain audio, avoid capturing while conflicting record/overdub actions are running, and confirm the source is audible in RUN4.
- MIDI does not appear: use Settings > MIDI > Refresh Devices, choose Input All, check the source app/device channel, or connect through Bluetooth MIDI.
- Hardware keyboard does not play Poly: select Poly, press M or tap Keys to enable QWERTY MIDI, and leave any active text field. With Keys On, A through K are notes rather than navigation shortcuts.
- AUv3 instrument does not load or reports that it could not connect safely: RUN4 automatically retries one recoverable graph connection. If it still fails, refresh the AUv3 list, confirm the plugin supports a Music Device component, then remove and reload it.
- A Docked or Compact AUv3 editor is blank: close and reopen the editor or try Full Screen. Some plugins advertise only specific view configurations even though their audio engine is running.

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- A tempo-based AUv3 does not follow transport: confirm RUN4 transport is running and the plugin's own sync/host-transport option is enabled.
 - Audio dropouts: reduce heavy AUv3 plugins, lower dense Grain/Bloom/Command activity, stop unused tracks, freeze/bounce complex material, or export stems.
 - A linked file is missing: open Samples, check Linked Folders, re-link the folder or import the file again.
 - Offline bounce has disabled stems: read the detail line next to the track. Some live or plugin-driven states may require live export instead.